

# MATA37H3 — Week 02

## *The Fundamental Theorem of Calculus*

### Recap and orientation

Unit 7 defined  $\int_a^b f(x) dx$  as a limit of Riemann sums — a definition that is rigorous but nearly impossible to compute directly for anything but the simplest functions. This unit introduces the **Fundamental Theorem of Calculus (FTC)**, which converts definite-integral computation into an *antiderivative-lookup* problem: no more limits of sums needed. FTC has two parts — Part 1 says differentiating an integral gets you back the original function; Part 2 says you can evaluate a definite integral by plugging endpoints into any antiderivative. Together they say, informally, “integration and differentiation undo each other.”

### Why this matters

Every integral you compute from Unit 9 onward — by substitution, by parts, with partial fractions — is really just “find an antiderivative, then invoke FTC Part 2.” Misunderstanding FTC is also one of the most common sources of subtle errors on rigorous exams: confusing “an antiderivative” with “the antiderivative  $\int_0^x f$ ”, forgetting the chain rule when the upper limit is itself a function of  $x$ , and misapplying FTC to functions that are only piecewise continuous.

#### DEFINITION Antiderivative and Indefinite Integral (MAT137 Unit 8)

A function  $F$  is an **antiderivative** of  $f$  on an interval  $I$  if  $F'(x) = f(x)$  for every  $x \in I$ . If  $F$  is one antiderivative of  $f$  on  $I$ , then *every* antiderivative of  $f$  on  $I$  has the form  $F(x) + C$  for some constant  $C$  — this is why we write  $\int f(x) dx = F(x) + C$  (the **indefinite integral**), with the understanding that  $C$  ranges over all real numbers.

**Watch for:** the “ $+C$ ” fact requires  $I$  to be a *single interval*. On a domain with a gap (like  $\mathbb{R} \setminus \{0\}$ ), two antiderivatives can differ by a *different* constant on each piece — this is exactly the trap in Worked Example 1 below.

#### KEY FACT Functions Defined by an Integral (MAT137 Unit 8)

An expression like  $F(x) = \int_a^x f(t) dt$  (with  $x$  as the *upper limit*, and a dummy variable  $t$  inside) defines a genuine function of  $x$ : for each fixed  $x$ , the integral is a single number. This is a completely valid way to define a function, **even when  $f$  has no elementary antiderivative** (e.g.  $F(x) = \int_0^x e^{-t^2} dt$ ). It is *not* valid to have the same free variable appear both as the bound of integration and as the integration variable, e.g.  $\int_0^x \frac{x}{1+t^8} dt$  mixes them; the variable of integration must be a dummy (like  $t$ ), never the same letter as the limit.

**KEY FACT Fundamental Theorem of Calculus, Part 1 (OpenStax §1.3, Theorem 1.4)**

If  $f$  is continuous on  $[a, b]$ , and  $F(x) = \int_a^x f(t) dt$ , then

$$F'(x) = f(x) \quad \text{for every } x \text{ in } [a, b].$$

In words: the derivative of “the integral of  $f$  from a fixed point to a variable point  $x$ ” is just  $f(x)$  itself. This is the precise sense in which integration and differentiation are inverse operations.

**KEY FACT FTC Part 1 with a Function in the Limit (Chain Rule Extension)**

If  $F(x) = \int_a^{g(x)} f(t) dt$  for a differentiable function  $g$ , then by FTC Part 1 combined with the chain rule,

$$F'(x) = f(g(x)) \cdot g'(x).$$

If the **lower** limit is the variable instead,  $\int_{g(x)}^b f(t) dt = -\int_b^{g(x)} f(t) dt$ , so differentiating

picks up an extra minus sign:  $\frac{d}{dx} \int_{g(x)}^b f(t) dt = -f(g(x)) g'(x)$ . If **both** limits depend on  $x$ , split the integral at any convenient constant point and differentiate each piece separately.

**KEY FACT Fundamental Theorem of Calculus, Part 2 — the Evaluation Theorem (OpenStax §1.3, Theorem 1.5)**

If  $f$  is continuous on  $[a, b]$  and  $F$  is **any** antiderivative of  $f$  on  $[a, b]$ , then

$$\int_a^b f(x) dx = F(b) - F(a),$$

often written  $F(x) \Big|_a^b$ . This is what makes definite integrals computable: find *any one* antiderivative, evaluate it at the two endpoints, subtract. It does not matter which antiderivative you pick — the arbitrary constant  $C$  always cancels in the subtraction.

**KEY FACT General Solution to  $G'(x) = f(x)$ ,  $G(x_0) = y_0$  (MAT137 Unit 8)**

If  $f$  is continuous on  $\mathbb{R}$  and we want the *unique* function  $G$  with  $G'(x) = f(x)$  **and** a prescribed value  $G(x_0) = y_0$ , the answer is

$$G(x) = y_0 + \int_{x_0}^x f(t) dt.$$

This single formula answers “which antiderivative?” — a bare antiderivative of  $f$  is only ever determined up to  $+C$ ; pinning down one specific value  $G(x_0) = y_0$  removes that ambiguity

and produces exactly this formula, not  $\int_0^x f(t) dt$  unless  $x_0 = 0$  and  $y_0 = 0$ .

**DEFINITION Mean Value Theorem for Integrals & Average Value (OpenStax §1.3, Theorem 1.3)**

If  $f$  is continuous on  $[a, b]$ , there exists at least one point  $c \in [a, b]$  such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x) dx.$$

The right-hand side,  $\frac{1}{b-a} \int_a^b f(x) dx$ , is called the **average value** of  $f$  on  $[a, b]$  — this theorem says a continuous function always *attains* its own average value somewhere on the interval.

## Worked Examples

### EXAMPLE 1

*The most misunderstood antiderivative*

Find  $\int \frac{1}{x} dx$ , and explain why  $F_1(x) = \ln x$  alone is not a correct general answer.

$F_1(x) = \ln x$  has domain  $(0, \infty)$  only, so it cannot be the antiderivative on all of  $\mathbb{R} \setminus \{0\}$ . Instead consider  $F_3(x) = \ln|x|$ : for  $x > 0$ ,  $F_3(x) = \ln x$  so  $F'_3(x) = \frac{1}{x}$ ; for  $x < 0$ ,  $F_3(x) = \ln(-x)$  so by the chain rule  $F'_3(x) = \frac{1}{-x} \cdot (-1) = \frac{1}{x}$ . So  $F'_3(x) = \frac{1}{x}$  on **both** pieces of the domain, giving

$$\int \frac{1}{x} dx = \ln|x| + C.$$

But since the domain  $\mathbb{R} \setminus \{0\}$  is *not* a single interval, the constant is allowed to be **different on each side of 0**: the most general antiderivative is actually

$$G(x) = \begin{cases} \ln|x| + C_1 & x > 0 \\ \ln|x| + C_2 & x < 0 \end{cases}$$

for independently chosen  $C_1, C_2$ . This is why  $F_4(x) = \ln|2x| = \ln|x| + \ln 2$  is *also* a valid antiderivative on  $x > 0$  — it just corresponds to  $C_1 = \ln 2$  — and does not contradict the “ $+C$ ” rule, because that rule only guarantees a single shared constant on a connected interval. ■

### EXAMPLE 2

*FTC Part 1, straightforward and with the chain rule*

Find  $F'(x)$  for (a)  $F(x) = \int_2^x \sin(t^3) dt$ , and (b)  $G(x) = \int_1^{x^2} \frac{\sin t}{t^2} dt$ .

(a) By FTC Part 1 directly (the upper limit is bare  $x$ ),  $F'(x) = \sin(x^3)$ .

(b) Here the upper limit is  $g(x) = x^2$ , not  $x$  itself, so we need the chain-rule extension: with

$$f(t) = \frac{\sin t}{t^2},$$

$$G'(x) = f(g(x)) \cdot g'(x) = \frac{\sin(x^2)}{(x^2)^2} \cdot 2x = \frac{2x \sin(x^2)}{x^4} = \frac{2 \sin(x^2)}{x^3}. \quad \blacksquare$$

### EXAMPLE 3

*FTC Part 1 with the variable in the lower limit and both limits*

Find  $F'(x)$  for  $F(x) = \int_{2x}^{x^2} \frac{1}{1+t^3} dt$ .

Split at a convenient constant, say  $t = 0$ :

$$F(x) = \int_{2x}^0 \frac{dt}{1+t^3} + \int_0^{x^2} \frac{dt}{1+t^3} = -\int_0^{2x} \frac{dt}{1+t^3} + \int_0^{x^2} \frac{dt}{1+t^3}.$$

Differentiate each piece with the chain rule (using  $f(t) = \frac{1}{1+t^3}$ ):

$$F'(x) = -f(2x) \cdot 2 + f(x^2) \cdot 2x = \frac{-2}{1+8x^3} + \frac{2x}{1+x^6}. \quad \blacksquare$$

#### EXAMPLE 4

*Reading FTC Part 1 off a piecewise-linear graph*

Let  $f$  be the piecewise-linear function with  $f(0) = 0$ , rising with slope 1 to  $f(1) = 1$ , constant at 1 on  $[1, 2]$ , falling with slope  $-1$  to  $f(3) = 0$  and continuing to  $f(4) = -1$ , then constant at  $-1$  on  $[4, 5]$ . Let  $F(x) = \int_0^x f(t) dt$ . Sketch  $F$  and describe  $F'$ .

Since  $F$  is an antiderivative of  $f$  by FTC Part 1,  $F' = f$  — so wherever  $f$  is *positive*,  $F$  is increasing; wherever  $f$  is *negative*,  $F$  is decreasing; and  $F$  has a local max/min exactly where  $f$  crosses zero. Computing  $F$  by accumulating signed area under  $f$ :  $F(0) = 0$ ; on  $[0, 1]$ , area of a triangle  $= \frac{1}{2}$ , so  $F(1) = \frac{1}{2}$ ; on  $[1, 2]$ , add a  $1 \times 1$  rectangle, so  $F(2) = \frac{3}{2}$ ; on  $[2, 3]$ ,  $f$  falls linearly from 1 to 0, adding another triangle of area  $\frac{1}{2}$ , so  $F(3) = 2$  (this is  $F$ 's **maximum**, since  $f$  changes from  $+$  to  $-$  there); on  $[3, 4]$ ,  $f$  is negative (a triangle from 0 down to  $-1$ ), subtracting  $\frac{1}{2}$ , so  $F(4) = \frac{3}{2}$ ; on  $[4, 5]$ ,  $f \equiv -1$ , subtracting 1 more, so  $F(5) = \frac{1}{2}$ . So  $F$  rises from 0 to a peak of 2 at  $x = 3$ , then falls to  $\frac{1}{2}$  at  $x = 5$  — and  $F'(x) = f(x)$  throughout, confirming FTC Part 1 graphically.  $\blacksquare$

#### EXAMPLE 5

*FTC Part 2 — basic evaluation*

Evaluate  $\int_1^3 (2x^2 - 4) dx$ .

An antiderivative of  $2x^2 - 4$  is  $F(x) = \frac{2x^3}{3} - 4x$  (any choice of  $C$  works; take  $C = 0$ ). By FTC Part 2,

$$\int_1^3 (2x^2 - 4) dx = F(3) - F(1) = \left(\frac{2(27)}{3} - 12\right) - \left(\frac{2}{3} - 4\right) = (18 - 12) - \left(\frac{2}{3} - 4\right) = 6 + \frac{10}{3} = \frac{28}{3}. \quad \blacksquare$$

#### EXAMPLE 6

*Area between a function and the  $x$ -axis, watching for sign changes*

Find the area of the region bounded by  $y = \sin x$  and the  $x$ -axis on  $[0, 2\pi]$ .

$\sin x \geq 0$  on  $[0, \pi]$  and  $\sin x \leq 0$  on  $[\pi, 2\pi]$ , so **naively** computing  $\int_0^{2\pi} \sin x dx$  would let the negative part cancel the positive part — giving the *net signed area*, not the true geometric area. Instead, split and take absolute values:

$$\text{Area} = \int_0^\pi \sin x dx + \left| \int_\pi^{2\pi} \sin x dx \right| = [-\cos x]_0^\pi + \left| [-\cos x]_\pi^{2\pi} \right| = (1 - (-1)) + |-1 - 1| = 2 + 2 = 4. \quad \blacksquare$$

(Direct check:  $\int_0^{2\pi} \sin x dx = -\cos(2\pi) + \cos 0 = -1 + 1 = 0$  — the signed total, which is *not* the area — confirming why the split was necessary.)

**EXAMPLE 7**

*Uniqueness of the solution to an IVP-style antiderivative problem*

Find the unique function  $H$  with  $H'(x) = e^{\sin x}$  for all  $x \in \mathbb{R}$  and  $H(1) = -2$ .

By the Key Fact on Page 1, with  $x_0 = 1$ ,  $y_0 = -2$ ,  $f(t) = e^{\sin t}$ :

$$H(x) = -2 + \int_1^x e^{\sin t} dt.$$

This is the *only* function satisfying both conditions: any other antiderivative of  $e^{\sin x}$  differs from  $\int_1^x e^{\sin t} dt$  by a constant  $C$ , and demanding  $H(1) = -2$  forces  $C = -2$  exactly (since  $\int_1^1 e^{\sin t} dt = 0$ ). Writing  $H(x) = \int_0^x e^{\sin t} dt - 2$  instead would be **wrong** unless the extra constant happens to work out — check: at  $x = 1$  that expression gives  $\int_0^1 e^{\sin t} dt - 2 \neq -2$  in general, so that guess fails the initial condition. ■

**EXAMPLE 8**

*Level: Foundational — FTC Part 2 with a polynomial*

Evaluate  $\int_0^2 (3x^2 - 2x + 1) dx$ .

**Step 1.** Find an antiderivative term by term:  $\int 3x^2 dx = x^3$ ,  $\int -2x dx = -x^2$ ,  $\int 1 dx = x$ , giving  $F(x) = x^3 - x^2 + x$  (check:  $F'(x) = 3x^2 - 2x + 1$ , matches the integrand ✓).

**Step 2.** By FTC Part 2,  $\int_0^2 (3x^2 - 2x + 1) dx = F(2) - F(0)$ .

**Step 3.** Compute the two endpoint values:  $F(2) = 2^3 - 2^2 + 2 = 8 - 4 + 2 = 6$ ;  $F(0) = 0 - 0 + 0 = 0$ .

**Step 4.** Subtract:  $6 - 0 = 6$ . ■

**EXAMPLE 9**

*Level: Intermediate — FTC Part 1, both limits are functions of  $x$*

Find  $F'(x)$  for  $F(x) = \int_x^{x^3} \cos(t^2) dt$ .

**Step 1.** Split the integral at a convenient constant,  $t = 0$ , so each piece has one variable limit and one fixed limit:

$$F(x) = \int_x^0 \cos(t^2) dt + \int_0^{x^3} \cos(t^2) dt = -\int_0^x \cos(t^2) dt + \int_0^{x^3} \cos(t^2) dt.$$

**Step 2.** Differentiate the first piece using FTC Part 1 directly (upper limit is bare  $x$ ), with  $f(t) = \cos(t^2)$ :

$$\frac{d}{dx} \left[ -\int_0^x f(t) dt \right] = -f(x) = -\cos(x^2).$$

**Step 3.** Differentiate the second piece using the chain-rule extension, with  $g(x) = x^3$ ,  $g'(x) = 3x^2$ :

$$\frac{d}{dx} \left[ \int_0^{x^3} f(t) dt \right] = f(g(x)) \cdot g'(x) = \cos((x^3)^2) \cdot 3x^2 = 3x^2 \cos(x^6).$$

**Step 4.** Add the two pieces:  $F'(x) = 3x^2 \cos(x^6) - \cos(x^2)$ . ■

**EXAMPLE 10**

*Level: Challenge — FTC with a non-elementary integrand, both limits variable*

Find  $F'(x)$  for  $F(x) = \int_{\sin x}^{x^2} e^{t^2} dt$ . (Note:  $e^{t^2}$  has no elementary antiderivative — FTC Part 1 does not need one.)

**Step 1.** Split at  $t = 0$ :  $F(x) = - \int_0^{\sin x} e^{t^2} dt + \int_0^{x^2} e^{t^2} dt$ , with  $f(t) = e^{t^2}$ .

**Step 2.** Differentiate the upper-limit piece,  $g(x) = x^2$ ,  $g'(x) = 2x$ :

$$\frac{d}{dx} \left[ \int_0^{x^2} f(t) dt \right] = f(x^2) \cdot 2x = 2x e^{(x^2)^2} = 2x e^{x^4}.$$

**Step 3.** Differentiate the lower-limit piece,  $h(x) = \sin x$ ,  $h'(x) = \cos x$ :

$$\frac{d}{dx} \left[ - \int_0^{\sin x} f(t) dt \right] = -f(\sin x) \cdot \cos x = -\cos x e^{\sin^2 x}.$$

**Step 4.** Combine:  $F'(x) = 2x e^{x^4} - \cos x e^{\sin^2 x}$ . ■

## Practice Problems

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*Read the hint and the caution before you start each problem — not after.*

### Antiderivatives & Indefinite Integrals

1. Compute  $\int (3x^8 - 18x^5 + 1) dx$  and  $\int \sqrt[3]{x} dx$  by “guess and check.”

**HINT** Use the power rule in reverse on each term, and rewrite  $\sqrt[3]{x}$  as  $x^{1/3}$  before integrating.

**CAUTION — before you solve** When the exponent is  $1/3$ , the new exponent after integrating is  $4/3$ , and you divide by  $4/3$  (i.e. multiply by  $3/4$ ) — don’t mishandle the fraction.

*Your working:*

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2. Compute  $\int \sec x \tan x \, dx$  and  $\int \frac{1}{x+5} \, dx$ .

**HINT** Recall the two separate ‘guess and check’ antiderivative facts for  $\sec x \tan x$  and for  $\frac{1}{x+5}$ .

**CAUTION — before you solve** Don’t confuse the antiderivative of  $\sec x \tan x$  with that of  $\tan x$  — they are different functions.

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3. Find the domain and derivative of  $F(x) = \ln |3x|$ , and confirm it is a valid antiderivative of  $\frac{1}{x}$  on  $x > 0$  by identifying the corresponding constant  $C$  in  $\ln |x| + C$ .

**HINT** Differentiate  $\ln |3x|$  with the chain rule to find  $F'(x)$ , then compare the result to  $\ln x + C$  using a log property.

**CAUTION — before you solve** Don’t forget the chain-rule factor from differentiating the inside function  $3x$  — it cancels with the 3 underneath.

*Your working:*

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## Functions Defined by Integrals & FTC Part 1

4. Which of these are *valid* ways to define a function  $F(x)$  (i.e. the dummy variable and

the free variable are used correctly)? (a)  $F(x) = \int_0^x \frac{t}{1+t^8} dt$ ; (b)  $F(x) = \int_0^x \frac{x}{1+t^8} dt$ ;  
(c)  $F(x) = \int_3^0 \frac{t}{1+x^2+t^8} dt$ .

**HINT** For each option, check whether the free variable  $x$  appears anywhere inside the integrand as a dummy-variable stand-in.

**CAUTION — before you solve** The outer variable  $x$  appearing somewhere in the integrand doesn't automatically make a definition invalid — only using it *as the variable of integration* does.

Your working:

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5. Find  $F'(x)$  for  $F(x) = \int_5^x \cos(t^2) dt$ .

**HINT** The upper limit here is bare  $x$ , so FTC Part 1 applies with no extra chain-rule factor.

**CAUTION — before you solve** Don't add an unnecessary chain-rule multiplier when the limit is exactly  $x$ , not a function of  $x$ .

Your working:

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6. Find  $F'(x)$  for  $F(x) = \int_x^7 \sin^3(\sqrt{t}) \, dt$ . (Careful with the variable in the *lower* limit.)

**HINT** Flip the limits first (introducing a minus sign) so the variable is in the upper position, then apply FTC Part 1.

**CAUTION — before you solve** Don't forget the sign flip that comes from having the variable in the *lower* limit.

*Your working:*

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7. Find  $F'(x)$  for  $F(x) = \int_{\sin x}^{e^x} \frac{t}{1+t^8} \, dt$ .

**HINT** Both limits depend on  $x$  here, so differentiate the upper-limit and lower-limit contributions separately.

**CAUTION — before you solve** Don't only differentiate with respect to the upper limit — the lower limit needs its own chain-rule term with a minus sign.

*Your working:*

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**8.** Let  $G$  be an antiderivative of a continuous function  $f$  with domain  $\mathbb{R}$ . Decide true or false, with a brief justification: (a)  $G(x) = \int_0^x f(t) dt$  must hold. (b) There exists  $C \in \mathbb{R}$  such that  $G(x) = C + \int_0^x f(t) dt$ .

**HINT** For (a), think of a specific antiderivative that doesn't equal the bare integral from 0. For (b), this is exactly the '+C' fact restated.

**CAUTION — before you solve** Don't assume every antiderivative must be expressible with lower limit exactly 0 — the constant can absorb any base point.

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## FTC Part 2 & Applications

**9.** Evaluate  $\int_0^1 (e^x + e^{-x} - \cos(2x)) dx$ .

**HINT** Antidifferentiate each term separately, being careful with the sign on  $e^{-x}$ .

**CAUTION — before you solve** The antiderivative of  $e^{-x}$  is  $-e^{-x}$ , not  $e^{-x}$  — watch the chain-rule sign.

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10. Evaluate  $\int_{\pi/4}^{\pi/3} \sec^2 x \, dx$ .

**HINT** Recall which trig function is the antiderivative of  $\sec^2 x$ .

**CAUTION — before you solve** Don't confuse the antiderivative of  $\sec^2 x$  (which is  $\tan x$ ) with the derivative of  $\sec x$  (which is  $\sec x \tan x$ ).

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11. Find the area of the bounded region enclosed between  $y = x^2 + 3$  and  $y = 3x + 1$ .

**HINT** Find where the two curves intersect first, then check which curve is on top on that interval before setting up the integral.

**CAUTION — before you solve** Don't integrate top-minus-bottom in the wrong order — check which function is larger on the interval, or you'll get a negative area.

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12. Find the average value of  $f(x) = 9 - x^2$  on  $[0, 3]$ , and find a value  $c \in [0, 3]$  where  $f(c)$  equals that average value (Mean Value Theorem for Integrals).

**HINT** Compute the average value with the formula  $\frac{1}{b-a} \int_a^b f(x) dx$ , then solve  $f(c) =$  that average for  $c$ .

**CAUTION — before you solve** Don't forget to divide by  $b - a$  — reporting the bare integral as the average value is a common shortcut error.

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### More Practice, Graded by Level

**13.** (Level: Foundational) Evaluate  $\int_{-1}^1 (4x^3 - 3x^2) dx$ .

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**14.** (Level: Intermediate) Find  $F'(x)$  for  $F(x) = \int_1^{\ln x} t^2 dt$  (for  $x > 0$ ).

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**15. (Level: Challenge)** Find the average value of  $f(x) = \sec^2 x$  on  $\left[0, \frac{\pi}{4}\right]$ , and find the exact value  $c$  in that interval where  $f(c)$  equals the average (Mean Value Theorem for Integrals).

*Your working:*

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